MATH MA'AM ACADEMY

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STANDARD VI • SCIENCE

UNIT 6: CHANGES AROUND US

Expanded revision notes based on the supplied textbook page, with additional concepts, everyday examples and important scientific facts for application-based preparation.

1. WHAT IS A CHANGE?

A change is any alteration in the size, shape, position, state, appearance, properties or composition of a substance or object. Changes occur continuously in nature and in our daily lives.

- Examples: ice melting, a seed germinating, a bulb glowing, iron rusting, food cooking and day changing into night.
- A change may involve only a change in appearance or state, or it may produce one or more new substances.

2. CLASSIFICATION OF CHANGES

Basis	Type 1	Type 2
Speed	Fast: occurs in a short time (bursting a balloon, switching on a bulb).	Slow: takes a long time (rusting, growth of a plant, ripening of fruit).
Reversibility	Reversible: can be changed back to the original state by suitable means (melting/freezing water).	Irreversible: cannot easily be returned to the original state by ordinary means (cooking food, burning paper).
Occurrence in time	Periodic: repeats at regular intervals (day and night, seasons, clock hands).	Non-periodic: does not repeat at regular intervals (earthquake, sudden landslide).
Nature of substance	Physical: no new substance is formed; composition remains the same (melting wax).	Chemical: new substance(s) with different properties are formed (rusting, burning).
Cause / source	Natural: occurs in nature without deliberate human action (earthquake, seasonal change).	Human-made: brought about deliberately by people (making cement, baking bread).
Effect on usefulness	Desirable: useful or wanted (cooking food, making curd).	Undesirable: harmful or unwanted (food spoilage, corrosion of a bridge).

Important: these classifications describe different aspects of a change. A single event can belong to several categories at once. For example, rusting is slow, chemical, generally irreversible, and often undesirable.

3. PHYSICAL CHANGES

A physical change changes the physical properties of a substance—such as shape, size or state—without forming a new substance. The chemical composition remains the same.

- Examples: melting ice, freezing water, boiling water, condensation of steam, dissolving sugar in water, cutting paper, breaking glass, grinding chalk and stretching a rubber band.
- Many physical changes are reversible, but not all. Cutting paper and breaking glass are physical changes even though the original object cannot easily be restored.
- A change of state is generally physical: ice, liquid water and water vapour are all H_2O.

Common changes of state

Process	Meaning	Example
Melting	Solid changes into liquid on absorbing heat.	Ice → water
Freezing / solidification	Liquid changes into solid on losing heat.	Water → ice
Vaporisation	Liquid changes into gas; includes evaporation and boiling.	Water → water vapour
Evaporation	Slow surface change from liquid to vapour; can occur below boiling point.	Wet clothes drying
Boiling	Rapid vaporisation throughout a liquid at its boiling point (for a given pressure).	Water boiling
Condensation	Gas or vapour changes into liquid on cooling.	Dew / droplets on a cold glass
Sublimation	Some solids change directly into gas without becoming liquid first.	Camphor, naphthalene, dry ice

Dissolving and mixtures

- When sugar or salt dissolves in water, no new substance is normally formed. The dissolved substance can often be recovered by evaporating the water; this is a physical change.

- A solution is a homogeneous mixture. Dissolving does not always mean a substance has chemically changed.

4. CHEMICAL CHANGES

A chemical change forms one or more new substances with properties different from those of the original materials. Chemical changes are often difficult to reverse by ordinary methods.

- Examples: burning wood, rusting iron, cooking food, curdling milk, digestion, fermentation, ripening of fruits and reaction of an acid with a base.
- Energy may be released or absorbed as heat, light, sound or electrical energy during a chemical reaction.

Common indicators of a chemical change

- Change in colour that is due to formation of a new substance.
- Evolution of a gas (not simply boiling or evaporation).
- Formation of a precipitate—a new insoluble solid in a solution.
- Release or absorption of heat or light, or a noticeable temperature change.
- Change in smell, such as food becoming spoiled.

No single sign alone proves a chemical change in every case. For example, bubbles can be produced by boiling (physical) or by a chemical reaction. Look for evidence that a new substance has formed.

Important chemical changes

Change	What happens / why it matters
Rusting	Iron reacts with oxygen and moisture to form hydrated iron oxides (rust). It weakens iron objects and structures.
Combustion	A substance reacts rapidly with oxygen, releasing heat and often light. Burning fuels is a chemical change.
Cooking	Heat causes chemical reactions in food, changing flavour, colour, texture and digestibility.
Fermentation	Microorganisms convert sugars into other substances. Yeast can convert sugar into ethanol and carbon dioxide.
Photosynthesis	Green plants use light energy to convert carbon dioxide and water into glucose and oxygen.
Respiration	Cells release usable energy from food molecules; aerobic respiration uses oxygen and forms carbon dioxide and water.
Digestion	Large food molecules are chemically broken into smaller molecules by enzymes and digestive juices.
Curd formation	Bacteria convert lactose in milk into lactic acid, causing milk proteins to coagulate.
Rancidity	Fats and oils react with oxygen or undergo other breakdown, producing unpleasant smell and taste.

5. REVERSIBLE AND IRREVERSIBLE CHANGES

- Reversible change: a change in which the original state or material can be obtained again by suitable means. Examples: melting and freezing of water, stretching a spring within its elastic limit, inflating and deflating a balloon.
- Irreversible change: a change in which the original material or object cannot be restored easily by ordinary means. Examples: burning paper, cooking an egg, setting cement and rusting iron.
- Reversibility and physical/chemical nature are not identical categories. Breaking glass is physical but practically irreversible; some chemical reactions can be reversed under special conditions.

6. PERIODIC AND NON-PERIODIC CHANGES

- Periodic: occurs repeatedly at regular intervals. Examples: day and night, phases of the Moon (approximately monthly), seasons, heartbeat and swinging of a pendulum (when maintained).
- Non-periodic: does not occur at fixed, regular intervals. Examples: earthquakes, volcanic eruptions, accidental breaking of a glass and sudden landslides.

A change that repeats is not necessarily perfectly identical each time. The important feature is its regular interval or predictable cycle.

7. FAST AND SLOW CHANGES

Fast changes	Slow changes
Bursting a balloon	Growth of a child
Lighting a matchstick	Rusting of iron
Breaking a glass	Ripening of a fruit
Switching on a bulb	Growth of nails and hair
Cutting a piece of paper	Formation of soil

The speed of a change depends on conditions. For example, iron rusts faster in moist air and may rust faster in salty conditions than in dry air.

8. NATURAL, HUMAN-MADE, DESIRABLE AND UNDESIRABLE CHANGES

- Natural changes: changes that occur in nature, such as rainfall, seasonal changes, erosion, earthquakes and growth of plants.
- Human-made changes: changes caused intentionally or as a result of human activities, such as manufacturing, cooking, construction and burning fuels.
- Desirable changes: changes that are useful or wanted, such as cooking food, ripening fruit and making medicines.
- Undesirable changes: changes that cause damage or loss, such as rusting of machinery, food spoilage, forest fires and corrosion of bridges.

9. EXOTHERMIC AND ENDOTHERMIC CHANGES

- Exothermic: releases energy to the surroundings, often as heat. Examples include combustion, respiration and some neutralisation reactions.
- Endothermic: absorbs energy from the surroundings. Examples include melting ice, evaporation and photosynthesis (which absorbs light energy).
- A physical or chemical change may involve energy transfer. Exothermic/endothermic describes energy flow, not whether a new substance forms.

10. CYCLIC AND NON-CYCLIC CHANGES

- Cyclic changes: follow a repeating sequence or cycle, such as the water cycle, life cycle of a butterfly and seasonal cycle.
- Non-cyclic changes: do not follow a regular repeating sequence, such as the breaking of a glass or a landslide.

Periodic and cyclic are related ideas, but not exactly the same in every scientific context. For school-level questions, focus on whether the event repeats in a regular sequence or at regular intervals.

11. RUSTING AND CORROSION

Corrosion is the gradual destruction of a metal due to reaction with its environment. Rusting is the corrosion of iron and iron-containing materials.

- Rusting generally requires both oxygen and water (moisture). Salt and acidic conditions can speed it up.
- Rust is hydrated iron(III) oxide; it is usually reddish-brown, flaky and porous.
- Rusting is a chemical change because a new substance is formed.

Methods of preventing rusting

- Painting or applying varnish creates a barrier between iron and air/moisture.
- Oiling or greasing prevents water and oxygen from reaching the metal surface.
- Galvanisation coats iron or steel with zinc.
- Electroplating coats a metal with another metal, such as chromium or tin, for protection or appearance.
- Alloying: stainless steel contains chromium and often nickel, which improve corrosion resistance.

12. USEFUL EVERYDAY EXAMPLES: PHYSICAL OR CHEMICAL?

Example	Type	Reason
Melting candle wax	Physical	Wax changes state; no new substance is formed during melting.
Burning candle wax vapour	Chemical	Wax vapour reacts with oxygen to form new substances.
Cutting vegetables	Physical	Only size and shape change.
Cooking vegetables	Chemical	New substances form and properties change.
Dissolving salt in water	Physical	Salt can be recovered by evaporation.
Rusting of a nail	Chemical	Iron forms rust.
Breaking a chalk piece	Physical	Only size and shape change.
Setting of cement	Chemical	Chemical reactions form a hard solid structure.
Drying wet clothes	Physical	Water evaporates; fabric remains chemically the same.
Milk turning sour	Chemical	Bacteria produce lactic acid and other substances.

13. IMPORTANT SCIENTIFIC FACTS

- A physical change does not necessarily mean a reversible change.
- A chemical change involves the formation of new substance(s), not merely a change in shape, size or state.
- Melting, boiling, freezing and condensation are physical changes.
- Burning, rusting, cooking, digestion and fermentation are chemical changes.
- A candle shows both types: melting of wax is physical; burning of wax is chemical.
- A change may be fast and chemical (explosion) or slow and chemical (rusting).
- A change may be slow and physical (erosion) or fast and physical (breaking glass).
- The law of conservation of mass states that in a closed system, total mass remains constant during a chemical reaction.

14. QUICK REVISION TABLE

Process / event	Classification
Ice melting	Physical, reversible, endothermic
Water freezing	Physical, reversible, exothermic
Water evaporating	Physical, generally endothermic
Steam condensing	Physical, exothermic
Iron rusting	Chemical, slow, generally irreversible, undesirable
Paper burning	Chemical, fast, irreversible, exothermic
Seed germination	Chemical changes occur; slow, generally irreversible
Earthquake	Natural, sudden, non-periodic
Day and night	Natural, periodic, cyclic
Cooking food	Chemical, generally irreversible, desirable
Breaking glass	Physical, fast, practically irreversible
Photosynthesis	Chemical, endothermic (uses light energy)

15. HOW TO SOLVE APPLICATION-BASED QUESTIONS

- First ask: has a new substance formed? If yes, it is chemical; if no, it is physical.
- Then classify the same event by speed, reversibility, timing, origin, usefulness and energy transfer. More than one label may apply.
- Do not assume every physical change is reversible or every chemical change releases heat.
- For mixed examples such as a candle, separate the processes: melting wax is physical; combustion is chemical.

Learn the definitions, but focus especially on comparing similar examples and identifying which classification applies to each change.